

Central-tensor (W) block-encoding comparison: four schemes

Parameters: N_{IP} (central bond) = 256

$N_k = k^3$ for k in [1, 2, 3, 4, 5, 6] (so N_k in [1, 8, 27, 64, 125, 216])

phase_bitsize $b = 32$

This isolates the CENTRAL block encoder of the Coulomb kernel W only -- the outer X-matrix block encodings are excluded. Each bloq is the central piece extracted from the corresponding full THC block encoding (C_{diag} / C_{inner}), so sizing and optimal_T propagation match the four-way full-system report.

1. Diagonal kernel: $\sum_{\{Q,I,J\}} W^Q_{\{IJ\}} |Q,I,J\rangle\langle Q,I,J|$ via QROAM $\rightarrow Ry \rightarrow QROAM^{\dagger}$ (the direct BE's central piece; W treated as diagonal in I,J).
2. SVD interferometer: $\sum_Q |Q\rangle\langle Q| (x) W^q$ via the block-unitary interferometer (exchange Unitary-Synthesis central piece).
3. Frobenius: $\sum_Q |Q\rangle\langle Q| (x) W^q$ via the Clader block-matrix scheme (BlockDiagonalClassicalMatrixBlockEncoding, $U_A = U_L^{\dagger} U_R$).
4. Reflection: $\sum_Q |Q\rangle\langle Q| (x) W^q$ via the Householder reflection isometry (ReflectionRectangularBlockEncoding, $n_{reflections} = N_{IP}$).

Tuning policies (per curve):

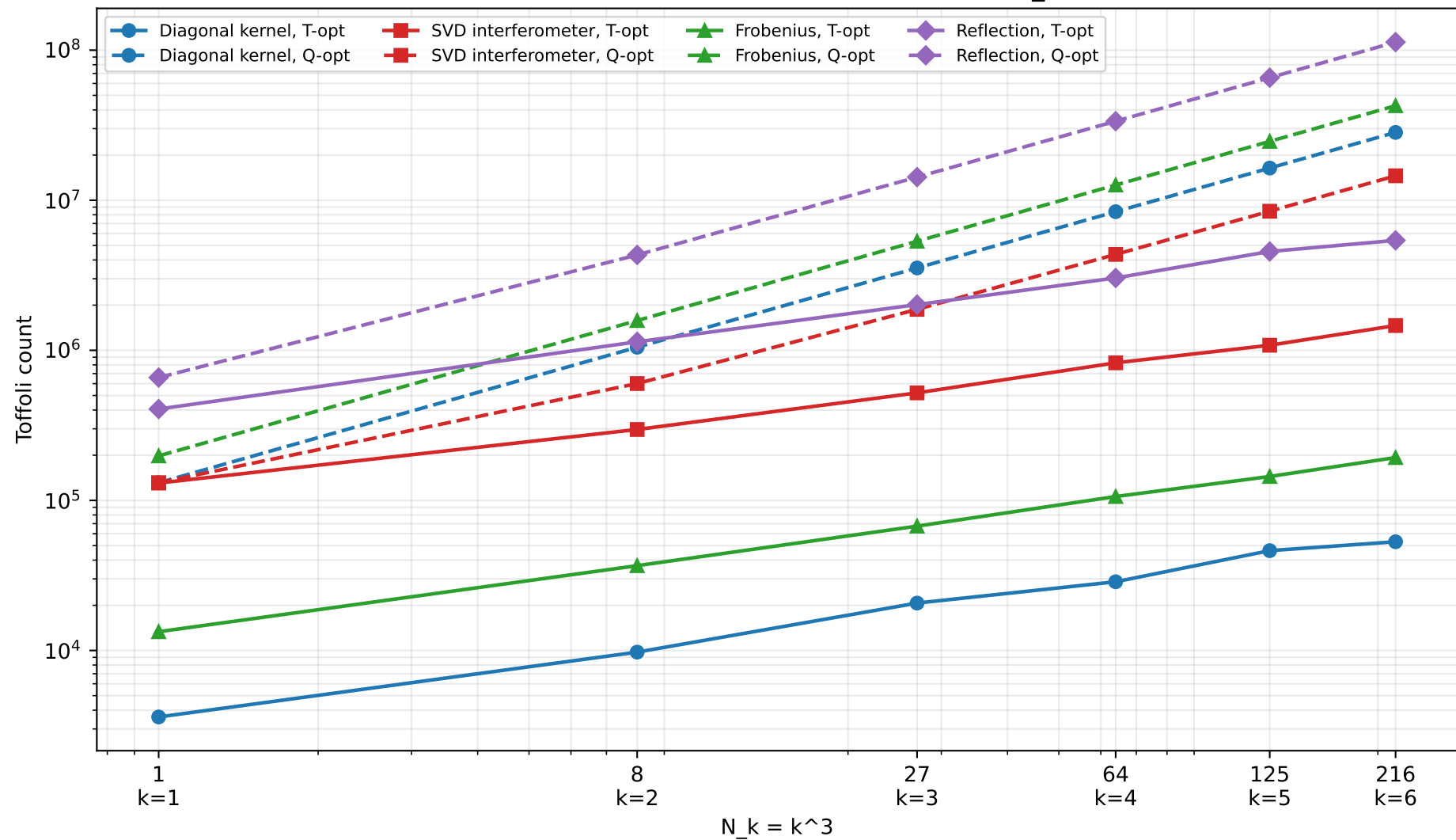
T-opt: every QROAM $\log_block_sizes$ at the closed-form Toffoli optimum.

Q-opt: no QROAM blocking ($\log_block_sizes = 0$) $\rightarrow$ minimal qubits.

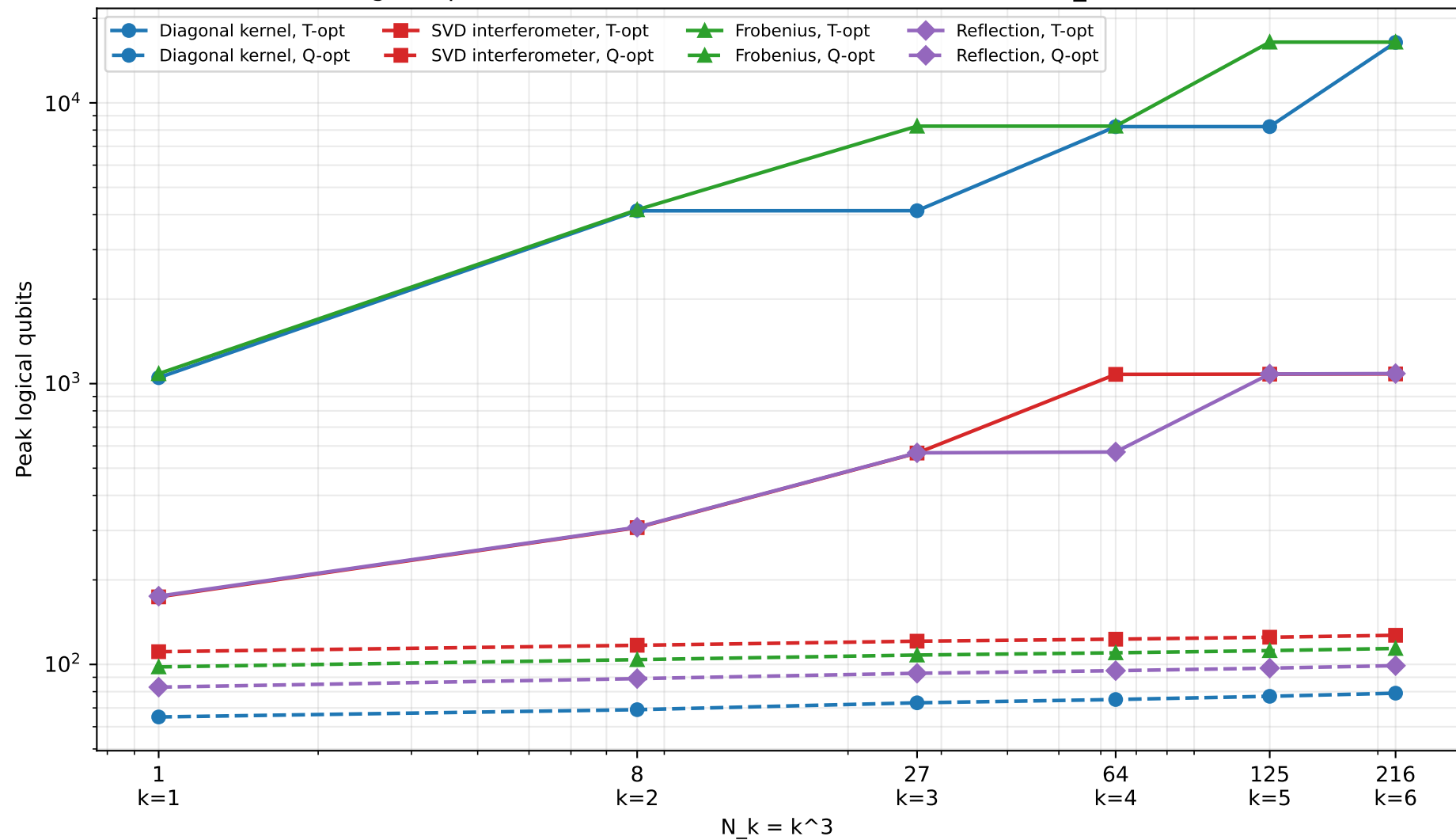
Note: schemes 1-3 (diagonal / Frobenius) trade many qubits for low T at T-opt; their Q-opt qubit counts grow only logarithmically in N_k .

All Toffoli / qubit counts come from Qualtran's `QECGatesCost` / `QubitCount`.

Toffoli count: central W block encoder, four schemes (N_IP=256, b=32)



Peak logical qubits: central W block encoder, four schemes (N_IP=256, b=32)



Toffoli count: central W block encoder, four schemes

central scheme	N_k=1	N_k=8	N_k=27	N_k=64	N_k=125	N_k=216
Diagonal kernel, T-opt	3,606	9,750	20,694	28,694	46,262	53,014
Diagonal kernel, Q-opt	131,127	1,048,631	3,538,999	8,388,663	16,384,055	28,311,607
SVD interferometer, T-opt	130,508	296,850	520,542	825,398	1,081,110	1,462,582
SVD interferometer, Q-opt	131,177	600,681	1,875,049	4,356,713	8,448,105	14,551,657
Frobenius, T-opt	13,339	36,739	67,383	106,116	144,331	192,945
Frobenius, Q-opt	197,938	1,579,590	5,329,791	12,632,814	24,672,933	42,634,422
Reflection, T-opt	406,272	1,139,968	2,016,512	3,034,880	4,555,008	5,403,904
Reflection, Q-opt	658,688	4,321,536	14,263,552	33,624,320	65,543,424	113,160,448

Peak logical qubits: central W block encoder, four schemes

central scheme	N_k=1	N_k=8	N_k=27	N_k=64	N_k=125	N_k=216
Diagonal kernel, T-opt	1,050	4,126	4,130	8,227	8,229	16,422
Diagonal kernel, Q-opt	65	69	73	75	77	79
SVD interferometer, T-opt	174	307	566	1,079	1,081	1,083
SVD interferometer, Q-opt	111	117	121	123	125	127
Frobenius, T-opt	1,085	4,161	8,260	8,262	16,455	16,457
Frobenius, Q-opt	98	104	108	110	112	114
Reflection, T-opt	175	308	567	571	1,082	1,086
Reflection, Q-opt	83	89	93	95	97	99